

School of Information Technologies
Faculty of Engineering & IT

ASSIGNMENT/PROJECT COVERSHEET - GROUP ASSESSMENT

Unit of Study: SOFT2412

Assignment name: **Group project Assignment 2 - Scrum Software Development**

Tutorial time: Every Wednesday from 16:00 to 18:00

Tutor name: **Ronald Noronha**

DECLARATION

We the undersigned declare that we have read and understood the *University of Sydney Student Plagiarism: Coursework Policy and Procedure*, and except where specifically acknowledged, the work contained in this assignment/project is our own work, and has not been copied from other sources or been previously submitted for award or assessment.

We understand that failure to comply with the *Student Plagiarism: Coursework Policy and Procedure* can lead to severe penalties as outlined under Chapter 8 of the *University of Sydney By-Law 1999* (as amended). These penalties may be imposed in cases where any significant portion of my submitted work has been copied without proper acknowledgement from other sources, including published works, the internet, existing programs, the work of other students, or work previously submitted for other awards or assessments.

We realise that we may be asked to identify those portions of the work contributed by each of us and required to demonstrate our individual knowledge of the relevant material by answering oral questions or by undertaking supplementary work, either written or in the laboratory, in order to arrive at the final assessment mark.

Project team members				
Student name	Student ID	Participated	Agree to share	Signature
1. Adam Behrmann	490612282	Yes	Yes	Adam
2. Lehan(Hans) Yang	500136245	Yes	Yes	Lehan Yang
3. Alex Tan	490379282	Yes	Yes	Alex
4. Shimiao Sun	480142841	Yes	Yes	Shimiao
5. Sung Bin Park	500574470	Yes	Yes	Sung Bin

Sprint 2 Report

Spring planning Thursday

Current sprint backlog

 BT-39

Update giftcard as 'redeemed' once it is used

 BT-38

Test cases for filtering movies

 BT-37

don't allow user to register for an account where the username already exists

 BT-35

complete loadCinemasFromDb() in Cinemas.java

 BT-34

complete loadMoviesFromDb() in Movies.java

 BT-33

As a customer, if I enter an incorrect cinema ID, I want to be given unlimited chances to try again

 BT-32

As a customer, if I enter an incorrect movie ID (when choosing a movie ID), I want to be given unlimited chances try ...

 BT-31

As a customer I want to be able to login to my account, so that I can view my bookings.

 BT-28

As a customer, I would like to book a ticket using a gift card that a friend has bought me

 BT-15

As a cinema manager, I want to add/remove cinema staff members, so I can control the employees of the cinema o...

 BT-13

As a user, I would like to have an option to remember my log in details, so I dont have to manually enter my passwo...

 BT-12

As a cinema staff, I want to gain the information about upcoming movies and show with their detailed data, so tha...

 BT-8

As a customer I want my booking transaction to be automatically cancelled after 2 minute timeout so that my secu...

 BT-7

As a customer I want to be able to cancel a booking so that I can change my mind.

 BT-5

As a customer I should be prompted to login/register (if i havent already) when attempting to book a movie so tha...

Product backlog

▼ Backlog (14 issues) 5 0 0 Create sprint

BT-42	As a cinema staff, I want to insert movie data for the upcoming week	5	TO DO ▼	
BT-43	As a cinema staff, I want to delete new data for the upcoming week		TO DO ▼	
BT-44	As a cinema staff, I want to modify movie for the upcoming week		TO DO ▼	
BT-45	As a cinema staff, I want to be able to check that, when a customer chooses to pay via gift card, the code they ente...		TO DO ▼	
BT-46	As a cinema staff, upon logged in, I want to see a list of the upcoming movies that include the item details		TO DO ▼	
BT-47	As a cinema staff, upon logged in, I want to see a summary that includes number of bookings for each movie sessio...		TO DO ▼	
BT-50	As a manager, upon logging in, I want to see a summary of cancelled transactions by users.		TO DO ▼	
✓ BT-51	list movies should only show movies for the upcoming week	5	TO DO ▼	
✓ BT-52	show the price of the booking, before we ask the user to confirm if they want to book		TO DO ▼	
✓ BT-53	Each customer should have a balance.		TO DO ▼	
BT-54	As a customer, I want to be able to check my balance after logging in		TO DO ▼	
✓ BT-55	Each gift card will have a balance.		TO DO ▼	
BT-56	As a customer, after logging in, I want to be able to redeem my gift card		TO DO ▼	
BT-57	As a customer, I want the option to top up my balance after my log in		TO DO ▼	

Standup Friday 22/10/2021 11:50

Attendees: all except Sung Bin

meeting minutes:

Adam :

- Will complete current tickets assigned, that are in progress
- Potentially blocked by code in book_movie_ticket branch not building. -> will be resolved

Alex :

- Has made use of outputStream to test parts of the program where system.in/system.out is used
- Will work on tickets PO has assigned

Hans:

- Will work on ticket assigned
- Will make a new sql server for testing

Sung Bin:

- absent

Shimiao:

- will start tickets

Standup Sunday 24/10/2021 13:00

Attendees: all

meeting minutes:

Adam :

- managed to pull mockito_test branch
- have 1 ticket working on BT 31
- To be completed by tonight (24/10/2021)
- BT 7 to be assigned to Alex

Alex :

- BT 5 was originally assigned to Adam, but I have completed it
- will do BT 7
- not doing BT-36, redundant ticket
- will delete redundant classes in mockito_tests branch
- I'll find some more tickets in backlog to work on

Hans:

- Set up a new sql server for testing use.
- Not yet started, will work on BT-5 and will find other free ticket after BT-5

Sung Bin:

- bt32, 33
- BT-28: related to gift card payments, since the current db doesn't support gift card trackings, should make a new table on the db to gain control
- If all issues are resolved, will work on password input masking

Shimiao:

- Find some more tickets from the backlog and finish them by Tuesday.
- BT12, BT15
- Will start StaffUI and implement BT12 and 15.

Standup Tuesday 26/10/2021 11:00 AM

Attendees: all

meeting minutes:

Adam

- Was tasked with completing BT31, code was reviewed in PR (https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/21)
- I will implement the changes in the review shortly after this meeting
- Will review tickets in backlog, and potentially assign more
- Will re-review backlog to ensure when all tickets are done, the assignment is completed

Alex

- continuing to work on bt 8 any idle activity that is more than 2 minutes (timeout) will automatically cancel the transaction. The cancelled transaction will return the application to the default page (list of movies).
- made changes/approved bt 13 and bt37
- requested changes for adam's branch
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/21

Shimiao

- Finished BT 12 and BT 15 for staff UI. changes can be reviewed in PR
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/18
- Created two table staffaccount and manageraccount in db.
- Will work on testcases tonight and try to finish them before second demo.

Hans

- Finished BT-13 and BT-37, fixed a major seat choosing bug in master branch. Code are reviewed in PR
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/22
- Solved PostgreSQL client environment on Jenkins server by migrating to physical environment from docker
- Will work on one more ticket if there are free tickets to assign
- Also will work on adding some testcases

Sung Bin

- Finished BT-28: added giftcards.sql, made relevant access functions to database.java, added UI actions to UserUI.java
- https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/20
- Will work on Booking function based on BT-32, 33, since the booking methods are clarified.

Sprint review

Feedback from client:

- It is redundant to allocate seats based on location (front middle rear) if we don't check for capacity of each area

- Instead of asking the user to type phrases e.g. “book movie” we can have the options represented as numbers to make it easier to type (more user friendly). It also reduces the chance the user inputs something incorrectly
- For card numbers -> See spec as it mentions it (card JSON file)
- For card numbers, each account can have an associated balance. When redeeming a gift card, $balance += card.balance$

Completed sprint tickets

 BT-49	As a manager, I want to be able to do everything a cinema staff member can do
 BT-48	As a manager, I want to be able to add/remove Cinema staff members.
 BT-41	The password input should be masked when the user types it
 BT-39	Update giftcard as 'redeemed' once it is used
 BT-38	Test cases for filtering movies
 BT-37	don't allow user to register for an account where the username already exists
 BT-35	complete loadCinemasFromDb() in Cinemas.java
 BT-34	complete loadMoviesFromDb() in Movies.java
 BT-33	As a customer, if I enter an incorrect cinema ID, I want to to be given unlimited ...
 BT-32	As a customer, if I enter an incorrect movie ID (when choosing a movie ID), I wa...
 BT-31	As a customer I want to be able to view my bookings.
 BT-28	As a customer, I would like to book a ticket using a gift card that a friend has b...
 BT-15	As a cinema manager, I want to add/remove cinema staff members, so I can co...
 BT-13	As a user, I would like to have an option to remember my log in details, so I do...
 BT-12	As a cinema staff, I want to gain the information about upcoming movies and s...
 BT-8	As a customer I want my booking transaction to be automatically cancelled aft...
 BT-7	As a customer I want to be able to cancel a booking so that I can change my mi...
 BT-5	As a customer I should be prompted to login/register (if i havent already) whe...

Sprint retro Thursday 28/10/2021

Alex

- **what went well** frequent pull requests was good; allowed us to make progress quickly
- very fast response over group messaging allowed quick communication
- Hans was quick in resolving environment related issues (creating a second db for testing, installing psql into the server, and modifying jacoco test coverage exclusions)
- **what didn't go well** many pull requests had merge conflicts with master, so even if they were approved, merge conflicts still had to be resolved
- a lot of failing tests;
- **what can be improved + how** when creating a PR, make sure your branch already includes the latest commits from master
- run tests locally and ensure all tests pass before committing and creating a PR

Sung Bin

- **What went well:** Simultaneous work on different branches, with different methods/functionalities. Everyone's work was integrated well at the end. Also, rapid feedback via groupchat amongst the groupmates allowed to easily clarify the points, leading to faster and more effective work.
- **What did not go well:** The number of testcases was lacking compared to the number of methods/functions. Personally, working speed was slow due to local environment issues, and low familiarity with java integrated SQL codes.
- **What can be improved & how:** Personally keep practice/studying java SQL. Also when making methods, testcases should also be made at the same time, or right after, so that all the methods could be tested whether they are working in a desirable way.

Shimiao

- **What went well:** finish user stories and tasks quickly and test cases; help check PR and provide improving measures; quick response on group chat and fix bugs.
- **What did not go well:** took lots of time on running sh scripts on windows system while doing test case; failed to match test case output format; did not test a function exhaustively and made a PR even it have bugs.
- **What can be improved & how:** test my codes more comprehensive before making a PR; review other's PR more frequently to make sure our master branch can be updated on time.

Hans

- **What went well:** This week we are more familiar with branch and merge staffs in git, the workflow becomes more efficient
- **What did not go well:** The test of code is not doing good, I used to commit the code which was not get test cases. Which will potentially cause a build crash on the CI end. Also this week I'm not very positive in the project, next week I'll contribute more to the project.
- **What can be improved & how:** The testcase before commit can be easily improved, I'll finish testcases for the new component and make sure it passed all the testcases I created and then commit. This will also increase our total test coverage since we are only have less than 70% coverage now.

Adam

- **what went well:** we were able to complete most of the functionality for customers, which is a sprint goal that we set. We were also able to begin working on the manager and cinema staff functionality, which means there is less work to do in the upcoming sprint
- We addressed the suggestions from the client from the previous demo, to show them a far better version of what will be the final product.
- **what didn't go well:** There may have been a misunderstanding on my part, which lead to me working from the wrong (outdated) branch initially. I still managed to get my part done for the sprint, but it could have been done sooner and with less confusion, had this not misunderstanding not taken place.;
- **what can be improved + how:** We need to improve the test case coverage, and we can do so by testing the new parts of the software we create, before creating a PR

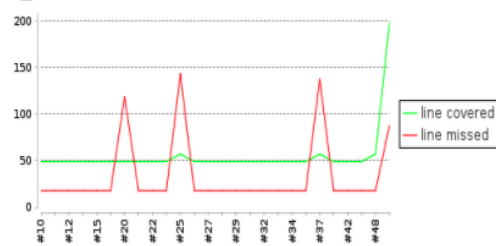
CI/CD

Jenkins:

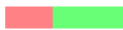
This time we faced a postgres sql environment in the previous jenkins, the sql client cannot be installed on the jenkins docker package, so the testcase cannot be operated, which caused all the testcases to fail on jenkins. So we migrated it to the physical server with PostgreSQL environment, now it can be operated properly. The link still <http://home.xsource.cc:8082/> . Also our testcases got a 78% line covered in the jacoco coverage report.

JaCoCo Coverage Report

 [Download jacoco.exec binary coverage file](#)



Overall Coverage Summary

name	instruction	branch	complexity	line	method	class
all classes	78%  M: 254 C: 889	65%  M: 33 C: 62	61%  M: 35 C: 54	69%  M: 88 C: 198	84%  M: 5 C: 26	100%  M: 0 C: 3

Coverage Breakdown by Package

name	instruction	branch	complexity	line	method	class
Assignment2_SOFT2412.ui	M: 254 C: 889 78% 	M: 33 C: 62 65% 	M: 35 C: 54 61% 	M: 88 C: 198 69% 	M: 5 C: 26 84% 	M: 0 C: 3 100% 

The issue with test processing:

Our team did commit untested code to their branch, which caused the build to fail on Jenkins. To solve this, we need to do tests locally, and make sure everything is working well then commit.

Individual contributions

Github repo https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412

- list all the Jira tickets you've been assigned and working on
- for the completed tickets, link the PR/commit/branch that you worked on that for that ticket

Adam

- Jira ticket BT 31 completed in this PR (yet to be merged):
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/21

Alex

- merged PR 24 and 25 into a new PR 26 and resolve conflicts
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/commit/38c576eb04a1cefd005ad2df6a4eb824df000327
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/26
- merge branches BT37 and BT13 together, then merge into master, resolving any merge conflicts https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/22
- Jira 38 test cases for movies
<https://soft2412.atlassian.net/jira/software/projects/BT/boards/1?selectedIssue=BT-38>
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/13/commits/393f57f784e42c608fe199c105027510ded1bd22
- Jira 7 <https://soft2412.atlassian.net/jira/software/projects/BT/boards/1?selectedIssue=BT-7>
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/17
- setup a framework to test our code since our code is console-based; abstracted System.out and System.in to use an abstract PrintStream and InputStream objects so that we could provide inputs as a String, and capture the printed contents as a String and compare it to the expected output (branch mockito_tests)

Hans

- Migrate Jenkins from docker to physical server to solve sql client environment
- Jira ticket BT-13 and BT-37
- BT-13: https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/22
- BT-37: https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/23
- Testcases and debug PR:
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/25

Shimiao

- Jira tickets BT12, BT15, BT49.
- BT15 and BT49 completed in branch Staff ui:
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/18
- BT12 completed in branch staff_merge:
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/26
and test cases PR:
https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/24

Sung Bin

- Jira tickets BT-28, BT-32, BT-33, BT-39
- Added giftcards table on database, added related functionality:
- https://github.sydney.edu.au/SOFT2412-2021S2/Assignment2_SOFT2412/pull/20

